ICS 604
Applied Data Science

Department of Information and Computer Sciences
University of Hawai`i at Mānoa

Kyungim Baek

Lecture 1

- Course Logistics
- Brief Introduction to Data Science

Instructor & teaching assistants

- Instructor: Kyungim (Kay) Baek
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  - Office hours: **Wednesdays 1:30 PM – 3:00 PM**, or by appointment
- Teaching assistants:
  - Ethan Chung (POST 303-B)
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  - Akib Sadmanee (POST 306-A)
    - Email: sadmanee (at) hawaii.edu
  - TA office hours will be held by appointment. You may also post questions on the course Slack channel.

Textbook

- There is no required textbook for this course. All class materials will be uploaded to Lamaku.
- Reference for data wrangling with Pandas:
  - McKinney, Wes. *Python for data analysis, 3rd Edition*. O'Reilly Media, Inc., 2022 (Open edition)

Grading criteria and exam schedule

- Homework assignments: **55%**
- Class participation: **10%**
- Final project: **35%**
  - Project proposal (5%)
  - Project proposal presentation (10%)
  - Final report and code (20 %)
    - **Due Monday, May 11**

Final grade

- Grading will follow standard percentage cutoffs:
  - 90% (A), 80% (B), 70% (C), and 60% (D), with a plus (+) assigned to the top 2% and a minus (–) to the bottom 2% within each range.
- Grading will not be on a curve.
- **No rounding** will be applied to the final points.

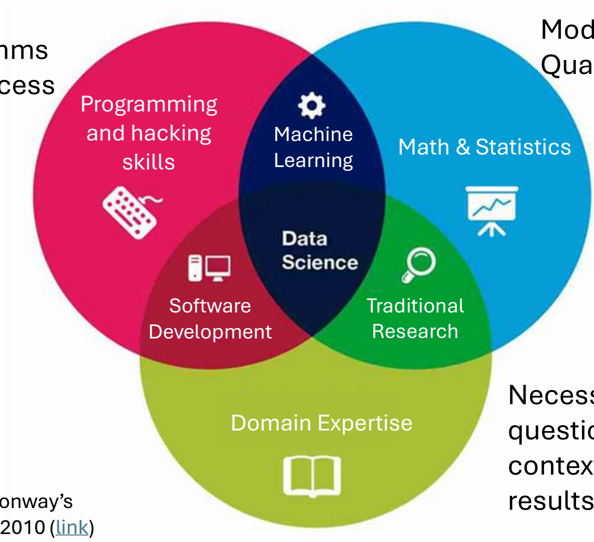
Course policies

- Homework assignments should be submitted electronically through Lamakū by **11:59 PM on the due date.**
- Email submissions will not be accepted unless Lamakū is unavailable at the time you submit your work.
- Late work is not accepted without prior approval, which is only granted in unavoidable circumstances; students should contact the instructor as soon as possible if any significant delays arise.
- There is no extra credit in this course.
- Review the syllabus for more detailed information about the course.

Questions?

What is data science?

Design and use algorithms to efficiently store, process and visualize the data



Model and summarize datasets; Quantify statistical uncertainty

Necessary to formulate the right questions, to put their answers in context, and to communicate results

Figure: Adapted from Drew Conway's Data Science Venn Diagram, 2010 (link)

What is data science?

- Aim: to transform raw data into actionable knowledge.
- Data science process can be described by the DIKW Pyramid.
  - A step of the pyramid adds value or answers a question by using its lower layers.

DIKW Pyramid: Data

- Data is the fundamental block of the DIKW pyramid and data science.
  - No data science without data.
- Data can arise from various processes. For example:
  - Manually collected or generated (e.g., survey responses, user photos).
  - Automatically generated (e.g., traffic sensors, IoT devices).
  - Simulated (e.g., weather or astronomy models).
- Raw data rarely has any intrinsic value for the entity that collects it.
  - Need to process it, interrogate and perhaps compare or enrich it with other datasets to generate extrinsic value.

DIKW Pyramid: Information

- The representation of the processed and organized data
  - Data cleaned (e.g., remove or impute missing values), disambiguated (e.g., fix alternative spellings), aggregated or enriched (e.g., link zip codes to median neighborhood revenue), etc.
- Exploratory data analysis (EDA) helps uncover information
  - Examples: visualizing distributions, computing summary statistics
    - E.g., mean daily usage of my app is ~1.5 hours.
- Information is useful but not directly actionable
  - It may not constitute valuable knowledge.
  - Does not directly lead to conclusions or decisions.

DIKW Pyramid: Knowledge

- Knowledge is the extraction of non-obvious insights from information
  - Focuses on how the information relates to our end-goals or decisions
- Examples of actionable insights
  - Are there differences in the usage of my app between males and females?
  - Which features of my app are strongly correlated with longer usage?
- Value of knowledge
  - In business or researcher, the knowledge you acquire provides a competitive advantage you are seeking to gain.

DIKW Pyramid: Wisdom

- Applying knowledge to make informed decisions and take actions.
- Knowledge are often uncertain, and resulting actions may not be optimal.
  - Conclusions are probabilistic and reasonable, leading to actions that are generally favorable.
- For example,
  - If feature X increases usage time, incorporating similar features is likely to further increase usage.
  - If feature Y is rarely used within the app, investing resources in similar features is unlikely to be effective or beneficial.
  - Although feature Z is associated with higher usage, its expensive implementation cost may limit its impact on revenue and overall value.

Python as a principal tool for data science

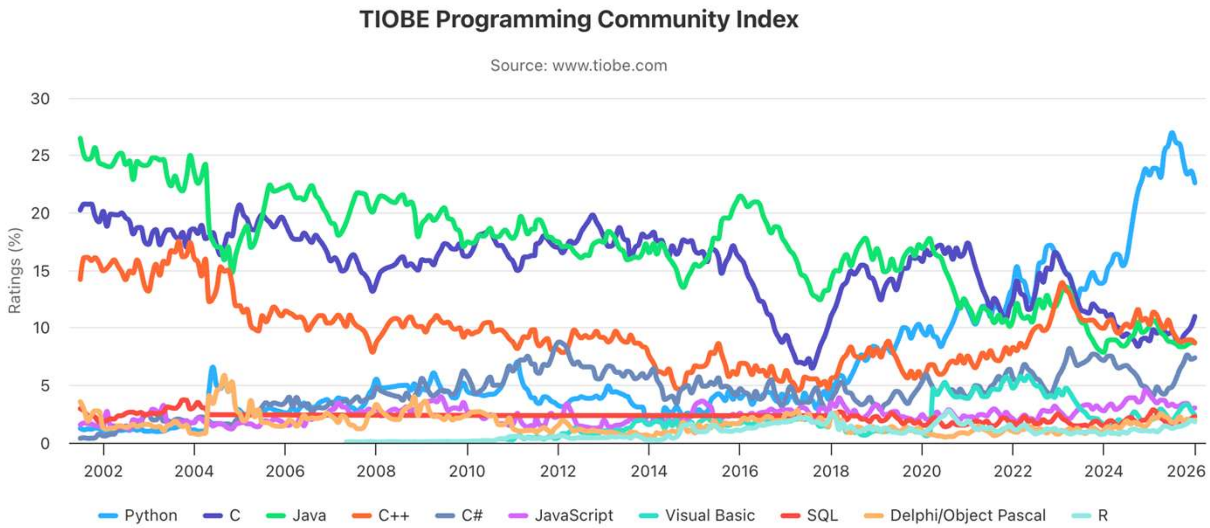
- Python is a general-purpose programming language.
- Supports rapid development of scripts and full-fledged applications.
  - You can write a quick analysis, a web portal (Django or Flask), or a business application quickly and (fairly) efficiently.
- A popular language with an abundance of resources.
  - Currently first in popularity according to Google Trends.
- Very simple syntax and gentle learning curve.

Python's popularity

Jan 2026	Jan 2025	Change	Programming Language	Ratings	Change
1	1		Python	22.61%	-0.68%
2	4	^	C	10.99%	+2.13%
3	3		Java	8.71%	-1.44%
4	2	v	C++	8.67%	-1.62%
5	5		C#	7.39%	+2.94%
6	6		JavaScript	3.03%	-1.17%
7	9	^	Visual Basic	2.41%	+0.04%
8	8		SQL	2.27%	-0.14%
9	11	^	Delphi/Object Pascal	1.98%	+0.19%
10	18	^^	R	1.82%	+0.81%

Source: https://www.tiobe.com/

Python’s popularity



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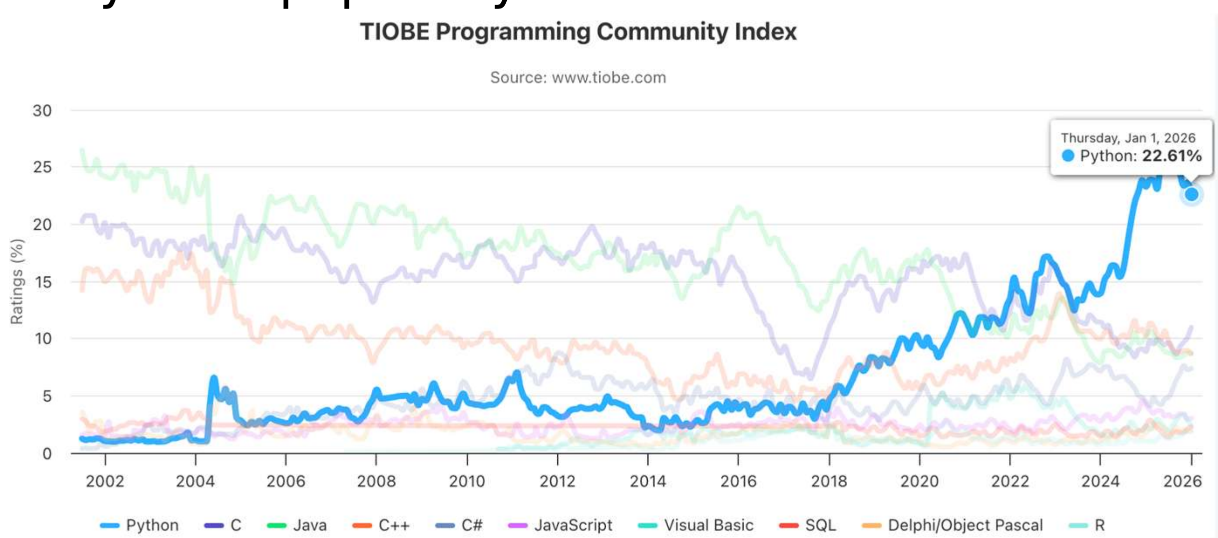
Python’s popularity



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Python's main advantages

- Gentle learning curve and easy-to-read syntax
- Supported on all major platforms, including low-cost devices like the $45 Raspberry Pi 5.
- Widely used across diverse fields.
  - Finance, biology, astronomy, oceanography, transportation, etc.
- Vast and active user community with a rich ecosystem of third-party packages.
  - Large collection of data science libraries spanning numerous domains.

Python covered in this course

- We will cover Python as a tool for data science.
  - Not interested in object-oriented programming or advanced features.
    - You should explore software engineering and advanced features in Python on your own.
- We will cover in depth functionality needed to deal with data at various stages. For example,
  - Manipulating data (e.g., read, clean, and normalize data)
  - Analyzing data (e.g., model data and test hypotheses)
  - Visualize data (e.g., plot distributions and inspect data)

Python versions

- We will be using Python 3 in this course
  - I am currently running Python 3.13
  - Make sure you are using Python > 3.10

```
from platform import python_version

python_version()
```

```
'3.13.5'
```

What we will cover

- The tools and materials we will cover will align with the DIKW pyramid.
  - **DATA**: Data wrangling and processing
  - **INFORMATION**: Exploratory data analysis, including data visualization and summary statistics
  - **KNOWLEDGE**: Modeling and machine learning
  - **WISDOM**: Model validation, hypothesis testing
- We will move back and forth and rotate between these topics.

What we will cover

- Data wrangling with Pandas
  - Covered primarily in the first part of the course.
- Data visualization with Matplotlib (with occasional use of Seaborn)
  - Covered throughout the course (with lighter emphasis since visualization is treated in more depth in another class).
- Probability distribution, summary statistics, parameter estimation and simulation
  - Explored using NumPy, SciPy, and StatsModels during the second and third quarters of the course.
- Introduction to machine learning models using StatsModels and Scikit-Learn
  - Covered in the final quarter of the course.

Interesting Python Resources

- How to Write Beautiful Python Code With PEP 8: https://realpython.com/python-pep8/
- Site with Excellent Python Tutorials and Demos : https://realpython.com
  - Many free tutorials + membership-based access

Assignments

- Set up environment:
  - Install Python and Jupyter Notebook
    - Install Python and necessary add-on packages separately, or
    - Install Anaconda distribution (https://www.anaconda.com/download), or
    - Use Miniconda (https://docs.anaconda.com/miniconda/miniconda-install/), and install individual packages that you use separately